InBody260S Open Protocol
User Guide

Version 3.2

Wednesday, Feburary 26, 2025

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1. Introduction

The host system can interface to InBody firmware through UART interface. The most common interfacing is done through the UART interface by using the Open Protocol (ASCII commands) that InBody firmware supports. With these Open Protocol (ASCII commands), the host can access InBody functionality without paying any attention to the complexity, which lies in the InBody protocol.

The user can write application code to the host processor with Open Protocol (ASCII commands). In this way, it is easy to develop InBody enabled applications.

2. Getting Started

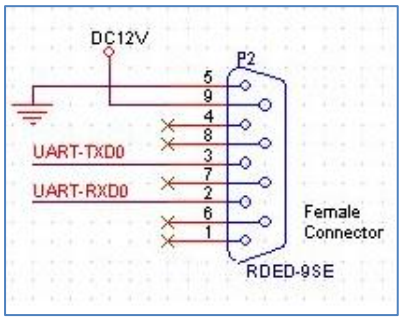
To start using InBody Open Protocol, you can use terminal software such as HyperTerminal.

2-1) Serial or USB Port

When using the terminal software, make sure that the InBody is connected to your PC's serial port. By default, InBody uses the following UART settings:

- Baud rate : 19200 bps
- Data Bits : 8 bit
- Parity Bits : No parity
- Stop Bits : 1bit
- Flow Control : None

- Serial Pin Description



2 : RXD
3 : TXD
5 : GND
9 : VCC (12V)
ETC : N.C

2-2) Lan or Wi-Fi

External device(Master) can communicate with InBody(Client) through TCP/IP.
External device(Master) can connect InBody(Client) when the same IP address is entered and port number is entered as '2004'.

2-3) Bluetooth

Refer to the Bluetooth manual additionally attached.

3. Open Protocol Structure

InBody can be used and controlled from the host system by sending Open Protocol (ASCII commands) through the UART interface to InBody. When installed and configured, the module can be commanded from the host with the following Open Protocol (ASCII commands):

3.1 Structure

STX	COMMAND	DATA	ETX

3.2 Structure Details

Parameter	Value	Size	Short Description
STX	0x02	1 Byte	Start of Frame
COMMNAD	0x63	1 Byte	Request InBody Status
	0x75	1 Byte	Send Personal Information Data
	0x64	1 Byte	Request Measured Data
	0x6C	1 Byte	Set "Lock Mode" / Release "Lock Mode"
	0x73	1 Byte	Skip the notice page in case of error in test progress
	0x69	1 Byte	Send Personal ID
	0x6B	1 Byte	Request Weight and Status
	0x77	1 Byte	Request Personal ID
	0x74	1 Byte	Set Time
DATA	-	x Byte	Data
ETX	0x03	1byte	End of Frame

NOTE :

- When transferring strings in DATA Parameter, nByte(string) and 0x1B(String Separater) will be sent.
- Typical Data Transmission Delay : 400-500ms
- Uses Half-Duplex communication method.

4. Technical Details

4.1 Command Listing

Command	Short Description
STATUS	Request InBody Status(Wait, Input, Measuring, Measure End, Lock)
INFO	Send Personal Information Data
RESULT	Request Measure Result Data
LOCK/UNLOCK	Set "Lock Mode" / Release "Lock Mode"

SKIP	Skip the notice page in case of error in test progress
SEND ID	Send Personal ID Data
WEIGHT	Request Weight and Status(Wait, Measuring, Measure End)
REQUEST ID	Request Personal ID Data
SET TIME	Set Time

NOTE :

- InBody only responses according to inputted command.

4.2 STATUS Command Description

Command "STATUS" is used to check that InBody status.

4.2.1 Syntax

Request (Host to InBody)
02 63 03

Response (InBody to Host)
02 63 [DATA] 03

Description		
DATA	30	Ready for measurement status
	31	After weight measurement, input personal information status
	32	Measuring status
	33	Measure completed status
	34	Lock status (ex> AutoLock)
	35	Error status (gender error: 1, height error: 2, age error: 3)

4.2.2 Example

(1) Ready for measurement status

Request (Host to InBody)
02 63 03
Response (InBody to Host)
02 63 30 1B 03

(2) After weight measurement, input personal information status

Request (Host to InBody)
02 63 03

Response (InBody to Host)
02 63 31 1B 03

(3) Measuring status

Request (Host to InBody)
02 63 03
Response (InBody to Host)
02 63 32 1B 03

(4) Measure completed status

Request (Host to InBody)
02 63 03
Response (InBody to Host)
02 63 33 1B 03

(5) Lock status

Request (Host to InBody)
02 63 03
Response (InBody to Host)
02 63 34 1B 03

(6) Error status

Request (Host to InBody)
02 63 03
Response (InBody to Host)
02 63 35 1B 31 1B 03 – Gender Error
02 63 35 1B 32 1B 03 – Height Error
02 63 35 1B 33 1B 03 – Age Error

4.3 INFO Command Description

Command "INFO" is used to send personal information data.

4.3.1 Syntax

Request (Host to InBody)
02 75 [DATA] 03

Description		
DATA	ID	ID Data. Max 14bytes.
	Gender	Gender Data. 1byte. ('M' or 'F')

	Height	Height Data. Max 5bytes.
	Age	Age Data. Max 4bytes.
Response (InBody to Host)		
02 75 03		

4.3.2 Example

Request (Host to InBody)
02 75 4A 4F 48 4E 1B 4D 1B 31 37 33 2E 31 1B 32 38 2E 30 1B 03
Response (InBody to Host)
02 75 03

4.4 RESULT Command Description

Command "RESULT" is used to response InBody Measured data.

After the 'measurement completed'[02 63 33 1B 03], command [02 64 03] should be sent.

4.4.1 Syntax

Request (Host to InBody)
02 64 03

Response (InBody to Host)
02 64 [DATA] 03

Description		
DATA	-	InBody Measured Data. X Bytes. See the Table below for details

DATA NUM	**InBody260S Overseas (except USA)**
0	InBody ID
1	ID
2	Gender
3	Height
4	Age
5	Weight
6	Test Date

7	Test Time
8~14	None
15	Protein
16	Minerals
17	BFM (Body Fat Mass)
18	TBW(Total Body Water)
19	FFM(Fat Free Mass) LBM(Lean Body Mass)
20	SMM(Skeletal Muscle Mass)
21	None
22	BMI (Body Mass Index)
23	PBF(Percent Body Fat)
24	WHR (Waist-Hip Ratio)
25	None
26	None
27	BMR(Basal Metabolic Rate)
28~30	None
31	None
32	None
33	None
34	Mobile Number
35	None
36	VFL (Visceral Fat Level)
37~71	None
72	None
73	None
74	None
75	None
76	None

77~102	None
103	FFM of Right Arm LBM of Right Arm
104	FFM of Left Arm LBM of Left Arm
105	FFM of Trunk LBM of Trunk
106	FFM of Right Leg LBM of Right Leg
107	FFM of Left Leg LBM of Left Leg
108	BFM of Right Arm
109	BFM of Left Arm
110	BFM of Trunk
111	BFM of Right Leg
112	BFM of Left Leg
113~129	None
130	None
131	None
132	None
133	None
134	None
135~146	None
147	Upper Limit (Protein Normal Range)
148	Lower Limit (Protein Normal Range)
149	Upper Limit (Minerals Normal Range)
150	Lower Limit (Minerals Normal Range)
151	None
152	None
153	Upper Limit (Weight Normal Range)
154	Lower Limit (Weight Normal Range)
155	Upper Limit (SMM Normal Range)

156	Lower Limit (SMM Normal Range)
157	Upper Limit (BMI Normal Range)
158	Lower Limit (BMI Normal Range)
159	Upper Limit (PBF Normal Range)
160	Lower Limit (PBF Normal Range)
161	Upper Limit (WHR Normal Range)
162	Lower Limit (WHR Normal Range)
163	Upper Limit (OD Normal Range)
164	Lower Limit (OD Normal Range)
165	None
166	None
167	None
168	None
169	Lower Limit (BFM Normal Range)
170	Upper Limit (BFM Normal Range)
171	Lower Limit (BMR Normal Range)
172	Upper Limit (BMR Normal Range)
173	Lower Limit (TBW Normal Range)
174	Upper Limit (TBW Normal Range)
175	Lower Limit (FFM Normal Range)
176	Upper Limit (FFM Normal Range)
177	None
178	None
179	None
180	None
181	Upper Limit (BMI Normal Range Next Value)
182~183	None

184	Nutrition Evaluation - Protein
185	Nutrition Evaluation – Minerals
186	Nutrition Evaluation – Body Fat
187~189	None
190	Obesity Evaluation - BMI
191	Obesity Evaluation - PBF
192	None
193	Body Balance Evaluation - Upper
194	Body Balance Evaluation - Lower
195	Body Balance Evaluation – Upper-Lower
196~202	None
203	Target Weight
204	Weight Control
205	BFM Control
206	FFM Control LBM Control
207	InBody Score
208~210	None
211	SMI (Skeletal Muscle Index)
212~230	None
231	None
232	None
233	None
234	None
235	None
236	None
237	None
238	None
239~305	None
306	Growth Score
307	Obesity Degree of a Child
308	Obesity Degree
309~312	None
313	recommended energy
314	Lean Mass Evaluation – Left Arm
315	Lean Mass Evaluation – Right Arm

316	Lean Mass Evaluation – Trunk
317	Lean Mass Evaluation – Left Leg
318	Lean Mass Evaluation – Right Leg
319	Body Fat Mass Evaluation – Left Arm
320	Body Fat Mass Evaluation – Right Arm
321	Body Fat Mass Evaluation – Trunk
322	Body Fat Mass Evaluation – Left Leg
323	Body Fat Mass Evaluation – Right Leg
324	None
325	None
326	None
327	None
328	None
329	None
330	None
331~347	None
348	None
349	None
350	None
351	None
352	None
353	None
354	None
355	None
356	None
357	None
358	None
359	None
360	None
361	None
362	None
363	None

364	None
365	None
366	None
367	None
368	None
369	None
370	None
371	None
372	None
373~485	None
486	None
487~743	None
744	Impedance Check
745~763	None

NOTE :

- None : 0x30

4.5 LOCK/UNLOCK Command Description

Command "LOCK/UNLOCK" is used to send lock/unlock command data.

* When InBody is set as Lock mode by this command, it cannot be unlocked unless InBody is initialized. So, the manual will be only provided upon (their) request.

4.5.1 Syntax

Request (Host to InBody)
02 6C [DATA] 03

Description		
DATA	Flag	0: Unlock (1 time) 1: Lock 2: Unlock

1. Once you LOCK the unit using this command, it will stay LOCKED
2. It can only be UNLOCKED momentarily using UNLOCK feature
3. Unit will default back to LOCKED status after completing the measurement or exiting out from Administrator Menu.

Response (InBody to Host)
02 6C 03

4.5.2 Example

Request (Host to InBody)
02 6C 31 1B 03
Response (InBody to Host)
02 6C 03

4.6 SKIP Command Description

Command "SKIP" is used to send skip command data.

4.6.1 Syntax

Request (Host to InBody)
02 73 03
Response (InBody to Host)
02 73 03

4.6.2 Example

Request (Host to InBody)
02 73 03
Response (InBody to Host)
02 73 03

4.7 Send ID Command Description

Command "Send ID" is used to send personal information data.

4.7.1 Syntax

Request (Host to InBody)
02 69 [DATA] 03

Description		
DATA	ID	ID Data. Max 14bytes.

Response (InBody to Host)
02 69 03

4.7.2 Example

Request (Host to InBody)
02 69 4A 4F 48 4E 1B 03
Response (InBody to Host)
02 69 03

4.8 WEIGHT Command Description

Command "WEIGHT" is used to check that InBody weight data.

4.8.1 Syntax

Request (Host to InBody)
02 6B 03

Response (InBody to Host)
02 6B [DATA] 03

Description		
DATA	Status	Status Data. Max 1bytes. (*See table below)

	Weight	Weight Data. Max 5bytes.
Description		
Status DATA	30	Ready for Weight measurement status
	31	Weight Measuring status
	32	Weight Measure completed status

4.8.2 Example

(1) Ready for Weight measurement status

Request (Host to InBody)
02 6B 03
Response (InBody to Host)
02 6B 30 1B 30 2E 30 1B 03

(2) Weight Measuring status

Request (Host to InBody)
02 6B 03
Response (InBody to Host)
02 6B 31 1B 35 38 2E 33 1B 03

(3) Weight Measure completed status

Request (Host to InBody)
02 6B 03
Response (InBody to Host)
02 6B 32 1B 36 33 2E 35 1B 03

4.9 Request ID Command Description

Command "Request ID" is used to request that User ID data.

4.9.1 Syntax

Request (Host to InBody)
02 77 03

Response (InBody to Host)
02 77 [DATA] 03

Description		
DATA	ID	ID Data. Max 14bytes.

4.9.2 Example

Request (Host to InBody)
02 77 03
Response (InBody to Host)
02 77 4A 4F 48 4E 1B 03

4.10 Set Time Command Description

Command "Set Time" is used to set time.

4.10.1 Syntax

Request (Host to InBody)
02 74 [DATA] 03

Description		
DATA	Time	Time Data. Max 19bytes. Ex> 2021:05:04/10:40:21

Response (InBody to Host)
02 74 03

4.10.2 Example

Request (Host to InBody)
02 74 32 30 32 31 3A 30 35 3A 30 34 2F 31 30 3A 34 30 3A 32 31 1B 03
Response (InBody to Host)
02 74 03

4.11 RESULT Command Description (Resend)

Command "RESULT" is used to response InBody Measured data.

After the 'measurement completed'[02 63 33 1B 03], command [02 64 03] should be sent.

4.11.1 Syntax

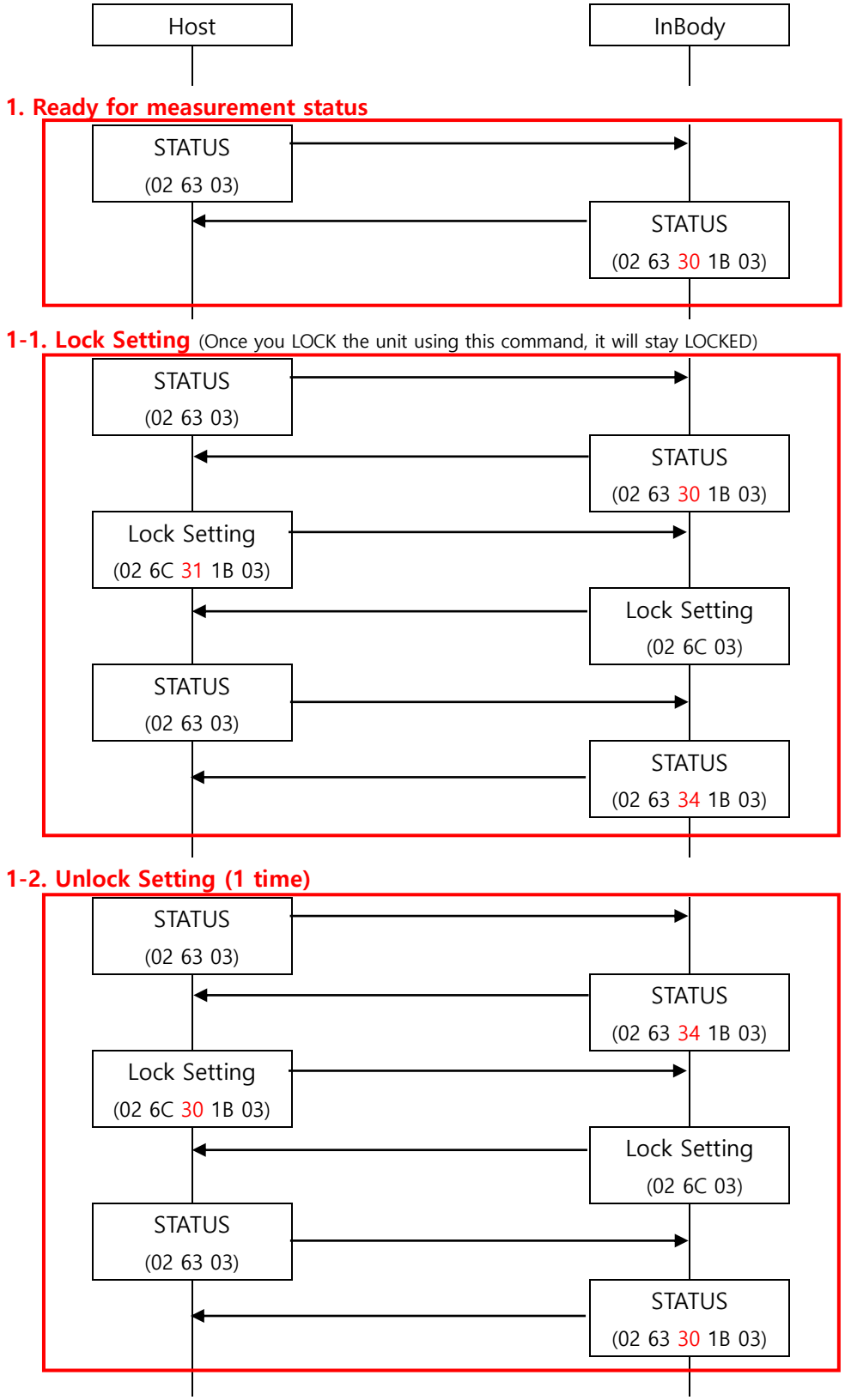
Request (Host to InBody)
02 64 03

Response (InBody to Host)
02 64 [DATA] 03

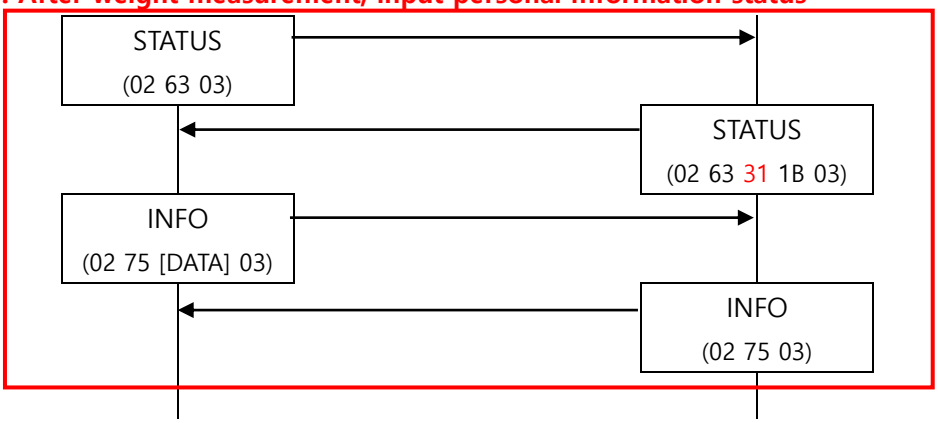
Description		
DATA	-	InBody Measured Data. X Bytes. See the Table below for details

5. Communication Example Figure

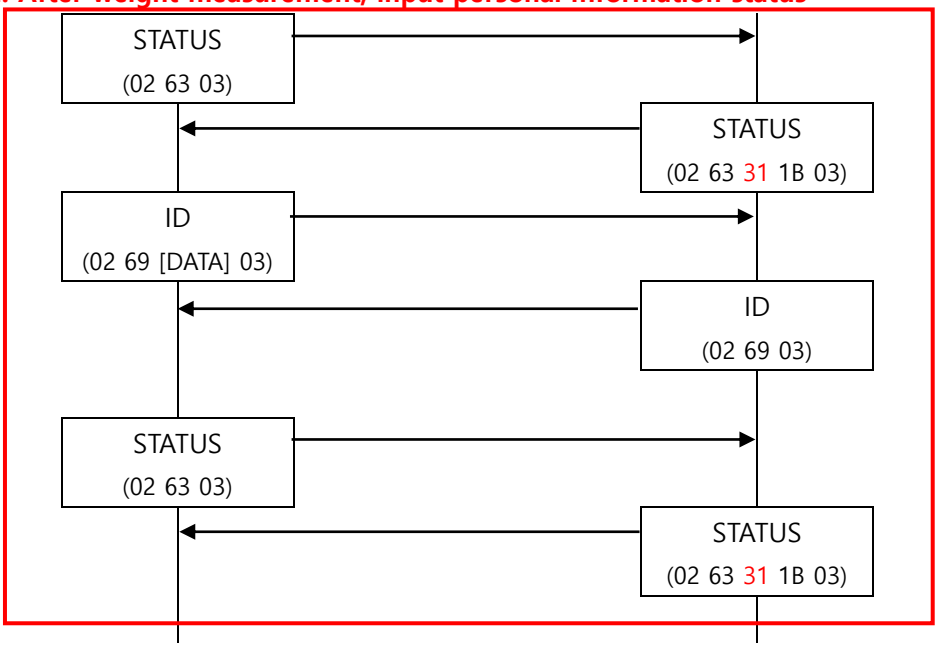
This example shows how to communicate Host and InBody.



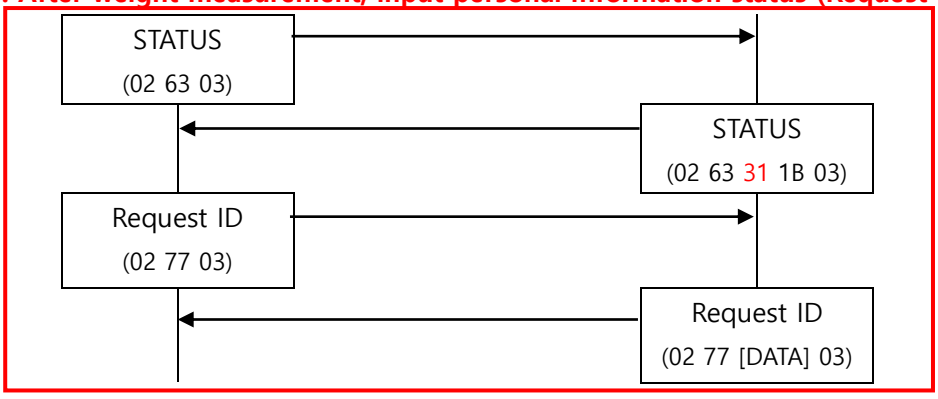
2-1. After weight measurement, input personal information status



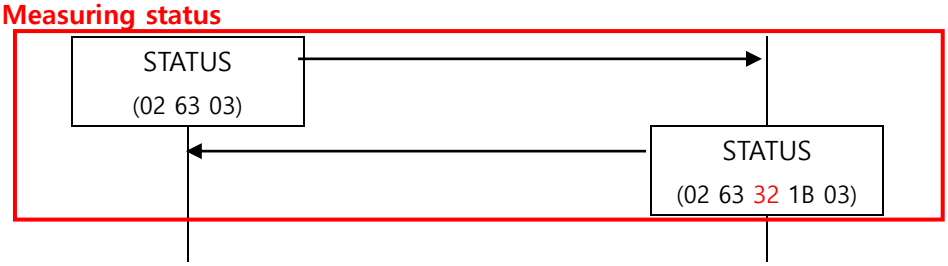
2-2. After weight measurement, input personal information status



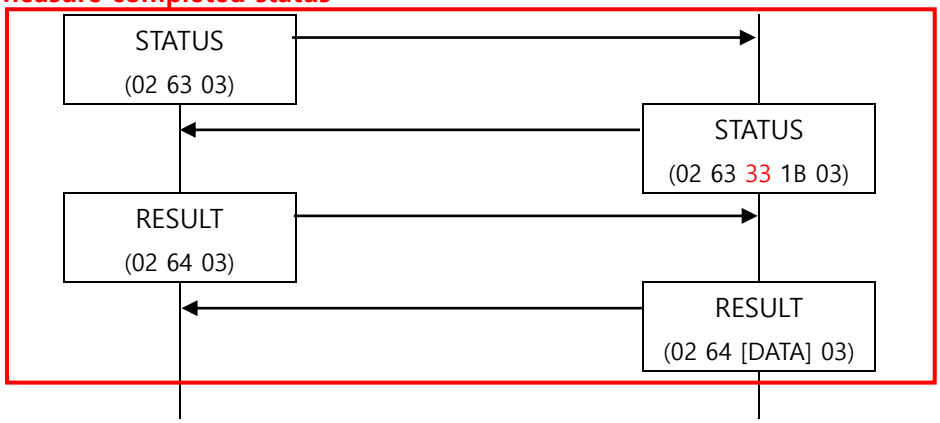
2-3. After weight measurement, input personal information status (Request ID)



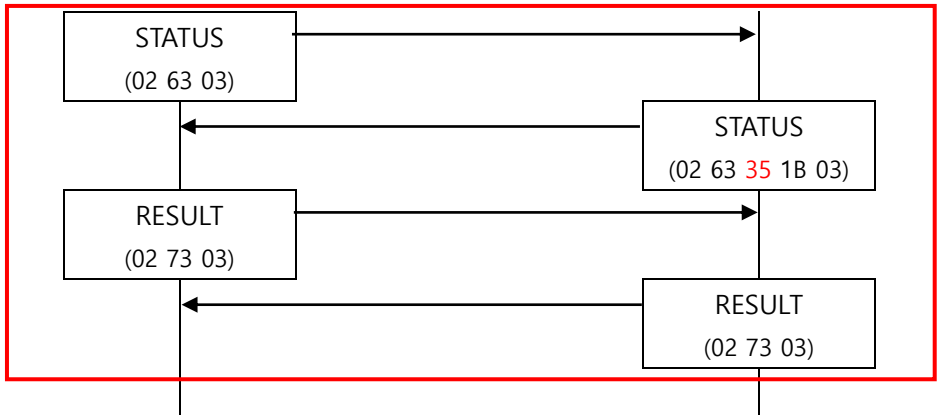
3. Measuring status



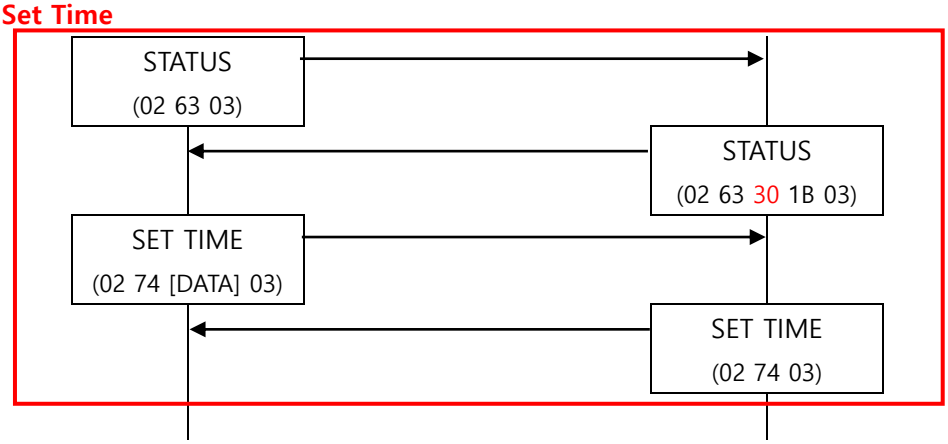
4. Measure completed status



5. Error Event status



6. Set Time



7. Resend status

